

Directivity and Beamforming in Antenna Arrays

PHY204 - Project P# 11

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Abstract. In this report, we analyse five half-wave dipole configurations, a single dipole, broadside and steered linear arrays, and broadside and steered planar arrays, using the array factor formalism, and apply them to optimise wireless coverage of the École Polytechnique campus via gradient ascent on a smoothed coverage–efficiency objective. The steered planar array proves optimal, covering 44 of 60 buildings and demonstrating that effective beamforming requires matching the radiation pattern to the spatial distribution of receivers rather than maximising raw directivity alone.

Code Repository : [link](#)

1 Introduction

This report investigates the radiation properties of five antenna configurations, all constructed from half-wave dipole elements, with increasing levels of complexity: from a single isolated dipole to a fully steerable two-dimensional planar array. The central questions are how the arrangement and phase control of multiple radiating elements shape the directional distribution of radiated power, and which configuration best serves a concrete engineering objective: providing wireless coverage across the buildings of the École Polytechnique campus.

The physical mechanism underlying all array configurations is wave interference. When multiple coherently fed antenna elements are arranged in space, their radiated fields superpose constructively in certain directions and destructively in others, producing a highly directional radiation pattern. The direction, width, and sidelobe structure of this pattern depend on the number of elements N , their inter-element spacing d , and the progressive phase shift δ applied between neighbouring elements. By varying these parameters electronically, without any mechanical movement, the main radiation lobe can be steered to any desired direction, a principle known as electronic beamforming, which underpins technologies from phased-array radar to 5G massive MIMO base stations.

This report draws on various sources of literature which are included in the bibliography. These sources include chapters from foundational textbooks on classical electrodynamics and antenna theory, notably Chapter 11 of Cadiz and Couairon’s *Classical Electrodynamics: Fundamentals and Applications* [1], which provides the theoretical framework for electromagnetic radiation, as well as Balanis’ *Antenna Theory: Analysis and Design* [2] and Ida’s *Engineering Electromagnetics* [3], both of which offer comprehensive treatments of antenna analysis and radiation patterns. In addition, technical documents and online resources were consulted to complement the theoretical content with practical considerations, including the *Antenna Patterns and Their Meaning* guide [4], the *Antenna-Theory.com* article on dipole antennas [5], the *Interference Technology* overview of antenna fundamentals [6], and the ETSI EN 300 328 glossary [7]. Chapters 9 and 10 on interference from Jérôme Faure’s *PHY-2F002_Waves, Optics and Radiation* course were also used to support the discussion on wave superposition phenomena. Finally, Anthropic’s *Claude Code* [8] was used as an AI-assisted tool for coding support in creating the optimisation algorithm.

To investigate these effects, this report employs the array factor formalism, in which the total radiation pattern factorises into a product of the single-element pattern $g(\vartheta)$ and the array factor $|\text{AF}(\vartheta, \varphi; \boldsymbol{\delta})|^2$.

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Starting from Maxwell's equations and the retarded potential, Section 2 derives the element factor of the half-wave dipole analytically, establishes the array factor for linear and planar configurations, and defines the directivity, half-power beamwidth, and sidelobe level for each. Section 3 describes the numerical implementation used to compute far-field patterns and directivities. Section 4 presents the simulation results for all five configurations and applies them to the campus coverage optimisation problem, in which a smooth coverage-efficiency objective J_k is maximised over the phase parameters of steerable configurations using gradient ascent with homotopy continuation.

The results confirm the theoretical scaling laws: beamwidth narrows as $1/N$, the first sidelobe level is universally -13.3 dB for uniform amplitude weighting, grating lobes appear when $d \geq \lambda$, and directivity degrades as $\sin \vartheta_0$ when the beam is steered away from broadside. For the campus optimisation, the steered planar array (Config. 5) achieves the highest combined coverage-efficiency score, covering 44 of 60 buildings after phase optimisation.

There exist many different types of antenna configurations, based not on dipoles, but also based on monopoles and patches. This report is therefore not an exhaustive evaluation of all antenna models, but rather an evaluation of increasingly complex configurations based on half-wave dipoles.

2 Theoretical background

2.1 Introduction and Physical Setup

We study five antenna configurations, all built from half-wave dipoles and all described by the same mathematical framework. Half-wave dipoles, meaning the physical length of each dipole is equal to half the operating wavelength $\lambda/2$ (this will be explained in section 2.2 below). The total radiation pattern of any configuration takes the form

$$U(\vartheta, \varphi) \propto g(\vartheta) \cdot |\text{AF}(\vartheta, \varphi; \boldsymbol{\delta})|^2,$$

where $g(\vartheta)$ is the half-wave dipole element factor, which describes the angular radiation pattern of an individual dipole element (i.e., how a single dipole radiates power as a function of direction), and AF encodes the array interference geometry, which describes the constructive and destructive interference between multiple elements due to their spatial arrangement and relative phase shifts.

Coordinate system. This report will use spherical coordinates (r, ϑ, φ) centered on the antenna feed point, with $\hat{\mathbf{z}}$ the array axis, $\vartheta \in [0, \pi]$ the polar (elevation) angle, and $\varphi \in [0, 2\pi)$ the azimuthal angle. All results hold in the far-field radiation zone $r \gg \lambda$.

Notation. $k = 2\pi/\lambda$ (wavenumber); I_0 (feed current); $\P_{in}$ (input power); N, N_x, N_z (number of elements); d, d_x, d_z (inter-element spacings); $\boldsymbol{\delta}$ (phase shift vector); $\eta = 1$ (antenna efficiency assumed to be an ideal lossless antenna throughout).

The five configurations. Five different configurations will be studied in this report, outlined below:

Table 1: The five configurations and their free parameters.

Config.	Description	Free parameters	$\boldsymbol{\delta}$
1	Single dipole	none	–
2	Linear, no steering	N, d	$\mathbf{0}$
3	Linear, steered	$N, d, \boldsymbol{\delta}$	free
4	Planar, no steering	N_x, N_z, d_x, d_z	$\mathbf{0}$
5	Planar, steered	$N_x, N_z, d_x, d_z, \boldsymbol{\delta}$	free

In order to describe the effectiveness of each configuration, we must describe the gain of the configuration, which is the ratio of radiation intensity in a given direction to the radiation intensity of an ideal isotropic radiator supplied with the same input power, and is given by

$$G^{(k)}(\vartheta, \varphi; \boldsymbol{\delta}) = D^{(k)} g(\vartheta) |\text{AF}^{(k)}(\vartheta, \varphi; \boldsymbol{\delta})|^2,$$

where $D^{(k)}$ is the configuration directivity, which describes how strongly the antenna concentrates radiated power in its most favorable direction compared to an isotropic source.

2.1.1 Radiation quantities

$$\langle \mathbf{S} \rangle = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \quad \text{power flux density [W m}^{-2}\text{]}$$

$$U(\vartheta, \varphi) = r^2 \langle S_r \rangle \quad \text{radiation intensity [W sr}^{-1}\text{]}$$

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^\pi U \sin \vartheta \, d\vartheta \, d\varphi \quad \text{total radiated power [W]}$$

$$\tilde{U}(\vartheta, \varphi) = \frac{U(\vartheta, \varphi)}{U_{\text{max}}} \in [0, 1] \quad \text{normalised radiation pattern}$$

$$D = \frac{U_{\text{max}}}{U_{\text{avg}}} = \frac{4\pi U_{\text{max}}}{P_{\text{rad}}} \geq 1 \quad \text{directivity}$$

$$G(\vartheta, \varphi) = \eta \frac{4\pi U(\vartheta, \varphi)}{P_{\text{in}}} = \eta D \tilde{U}(\vartheta, \varphi), \quad \eta = 1 \quad \text{power gain}$$

$$g(\vartheta) \equiv \tilde{U}_{\text{dipole}}(\vartheta) = \frac{\cos^2\left(\frac{\pi \cos \vartheta}{2}\right)}{\sin^2 \vartheta} \quad \text{element factor (half-wave dipole)}$$

$$\tilde{U}^{(k)}(\vartheta, \varphi; \boldsymbol{\delta}) = g(\vartheta) \cdot |\text{AF}^{(k)}(\vartheta, \varphi; \boldsymbol{\delta})|^2 \quad \text{pattern multiplication rule}$$

$$G^{(k)}(\vartheta, \varphi; \boldsymbol{\delta}) = D^{(k)} \cdot g(\vartheta) \cdot |\text{AF}^{(k)}(\vartheta, \varphi; \boldsymbol{\delta})|^2 \quad \text{master gain formula}$$

The total pattern of any configuration factorises into two independent contributions:

$$g(\vartheta) \equiv \tilde{U}_{\text{dipole}}(\vartheta) = \frac{U_{\text{dipole}}(\vartheta)}{U_{\text{dipole, max}}} \quad \text{element factor: normalised radiation pattern of a single isolated half-wave dipole}$$

$$\text{AF}^{(k)}(\vartheta, \varphi; \boldsymbol{\delta}) \equiv \frac{E_{\text{tot}}^{(k)}(\vartheta, \varphi; \boldsymbol{\delta})}{E_1(\vartheta, \varphi)} \quad \text{array factor: ratio of the total array field to the field of one isolated element; depends only on geometry (N, d) and phases (\boldsymbol{\delta})}$$

2.2 Radiation from a Dipole Antenna

Before studying how arrays of antennas combine to form directional beams, we must understand how a single antenna element radiates on its own. This section derives the far-field radiation pattern of the half-wave dipole, the element used in all five configurations, starting from Maxwell's equations in the Lorenz gauge and proceeding through the retarded potential. The central result is the *element factor* $g(\vartheta)$: the normalised radiation pattern of one isolated dipole, which will multiply the Array Factor in every subsequent configuration through the pattern multiplication rule $\tilde{U}^{(k)} = g(\vartheta) \cdot |\text{AF}^{(k)}|^2$. The derivation proceeds in three steps of increasing realism: the idealised Hertzian dipole ($\ell \ll \lambda$), the centre-fed wire of arbitrary length $2d$, and finally the half-wave special case $2d = \lambda/2$.

Retarded potential

In the Lorenz gauge, the vector potential of an arbitrary current distribution satisfies the wave equation $\nabla^2 \mathbf{A} - c^{-2} \partial_{tt} \mathbf{A} = -\mu_0 \mathbf{J}$, whose solution is the *retarded potential*:

$$\mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_{\text{ret}})}{|\mathbf{r} - \mathbf{r}'|} d^3 r', \quad t_{\text{ret}} = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}.$$

The retarded time t_{ret} encodes causality: the field at $(\mathbf{r}, t)$ is determined by the current at the earlier time t_{ret} , because the disturbance travels at speed c . The fields follow from $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\nabla V - \partial_t \mathbf{A}$. In the radiation zone $r \gg \lambda$, only r^{-1} terms survive, giving transverse fields with $\mathbf{B} = (\hat{\mathbf{r}} \times \mathbf{E})/c$.

Hertzian dipole

The *Hertzian dipole* is an infinitesimally short wire of length $\ell \ll \lambda$ carrying current $I_0 \cos(\omega t)$ along $\hat{\mathbf{z}}$. Since $\ell \ll \lambda$, the current is uniform and the retarded potential reduces to:

$$\mathbf{A} = \frac{\mu_0 I_0 \ell}{4\pi r} \cos(\omega t - kr) \hat{\mathbf{z}}, \quad k = \frac{2\pi}{\lambda}.$$

Applying the field relations and retaining only r^{-1} terms gives the far-field:

$$\mathbf{E} = -\frac{\mu_0 \omega^2 p_0}{4\pi c} \frac{\sin \vartheta}{r} \sin(\omega t - kr) \hat{\boldsymbol{\vartheta}}, \quad p_0 = \frac{I_0 \ell}{\omega}.$$

The radiated power per unit solid angle is:

$$U_{\text{Hertz}}(\vartheta) = \frac{\mu_0 \omega^4 p_0^2}{32\pi^2 c} \sin^2 \vartheta,$$

giving element factor $g_{\text{Hertz}} = \sin^2 \vartheta$: maximum at $\vartheta = \pi/2$, zero along the axis. This is the building block for the finite dipole.

Centre-fed dipole of arbitrary length

A wire of total length $2d$ along $\hat{\mathbf{z}}$, fed at $z = 0$ with current I_0 . Two approximations: (i) thin wire (radius $\ll \lambda$), and (ii) sinusoidal current distribution:

$$I(z, t) = I_0 \frac{\sin(k(d - |z|))}{\sin(kd)} \cos(\omega t), \quad |z| \leq d,$$

which satisfies $I(\pm d, t) = 0$. Treating the wire as a stack of Hertzian dipoles and superposing their contributions in the Fraunhofer approximation $r \gg d^2/\lambda$ — keeping the phase correction $r' \approx r - z' \cos \vartheta$ but setting $1/r' \approx 1/r$ in amplitude — gives:

$$\mathbf{E}(\mathbf{r}, t) = \frac{i\mu_0 c \sin \vartheta}{4\pi r} e^{i(kr - \omega t)} \int_{-d}^{+d} I(z') e^{-ikz' \cos \vartheta} dz' \hat{\boldsymbol{\vartheta}}.$$

Evaluating the integral:

$$U(\vartheta) = \frac{\mu_0 c I_0^2}{8\pi^2 \sin^2(kd)} \left(\frac{\cos(kd \cos \vartheta) - \cos(kd)}{\sin \vartheta} \right)^2.$$

This is the complete radiation pattern for any half-length d .

Half-wave dipole and element factor

Setting $d = \lambda/4$ gives $kd = \pi/2$, so $\cos(kd) = 0$ and $\sin(kd) = 1$. This is the half-wave dipole, which is the simplest resonant antenna, providing efficient radiation with good impedance (purely resistive input impedance $Z_{\text{in}} \approx 73 \Omega$). This is very widely used because of its predictable pattern and ease of use as a reference element in antenna configurations. The radiation pattern reduces to:

$$U^{(1)}(\vartheta) = \frac{\mu_0 c I_0^2}{2\pi^2} \cdot \frac{\cos^2\left(\frac{\pi \cos \vartheta}{2}\right)}{\sin^2 \vartheta}.$$

The angular dependence defines the *element factor of the half-wave dipole*:

$$g(\vartheta) = \frac{\cos^2\left(\frac{\pi \cos \vartheta}{2}\right)}{\sin^2 \vartheta}, \quad g(\pi/2) = 1. \quad (1)$$

This is the key result of this section. The directivity follows by integrating over the sphere:

$$D^{(1)} = \frac{4\pi U_{\max}^{(1)}}{P_{\text{rad}}} = 1.64 \quad (2.15 \text{ dBi}).$$

dBi describes decibels with respect to anisotropic radiator, meaning that the half-wave dipole radiates 2.15 dB more power in its strongest direction than a purely isotropic radiator. Both $g(\vartheta)$ and $g_{\text{Hertz}} = \sin^2 \vartheta$ vanish at $\vartheta = 0, \pi$ and peak at $\vartheta = \pi/2$. The half-wave dipole is slightly more directive, as shown by the HPBW (Half-Power Beamwidth), the angular width of the main radiation lobe, i.e. the angle between the directions at which the radiated power drops to half. (HPBW ≈ 78 vs 90). Crucially, $g(\vartheta)$ is φ -independent: the dipole is azimuthally symmetric and cannot form a directional beam in the horizontal plane; the fundamental limitation that motivates building arrays.

2.3 Single Half-Wave Dipole

The simplest configuration is a single half-wave dipole radiating in free space with no array structure. It has no phase variable and no Array Factor ($\text{AF}^{(1)} \equiv 1$), so the master gain formula reduces entirely to the element factor. It serves as the baseline against which all array configurations are compared.

Physical setup

A single half-wave dipole of total length $\lambda/2$ is oriented along $\hat{\mathbf{z}}$, fed at its midpoint with current amplitude I_0 . There are no free parameters: no spacing, no phase shift, no steering.

Radiation pattern and gain

The far-field radiation intensity is derived in Section 2.2 from the retarded potential. The result is:

$$U^{(1)}(\vartheta) = \frac{\mu_0 c I_0^2}{2\pi^2} \cdot \frac{\cos^2\left(\frac{\pi \cos \vartheta}{2}\right)}{\sin^2 \vartheta}.$$

Integrating over the full sphere gives the total radiated power P_{rad} , from which the directivity follows:

$$D^{(1)} = \frac{4\pi U_{\max}^{(1)}}{P_{\text{rad}}} = 1.64 \quad (2.15 \text{ dBi}).$$

The normalised pattern is exactly the element factor:

$$\tilde{U}^{(1)}(\vartheta) = g(\vartheta) = \frac{\cos^2\left(\frac{\pi \cos \vartheta}{2}\right)}{\sin^2 \vartheta}.$$

Applying the master gain formula with $\text{AF}^{(1)} \equiv 1$:

$$G^{(1)}(\vartheta) = D^{(1)} \cdot g(\vartheta) = 1.64 \cdot \frac{\cos^2\left(\frac{\pi \cos \vartheta}{2}\right)}{\sin^2 \vartheta}.$$

The pattern is azimuthally symmetric (φ -independent), vanishes along the wire axis ($\vartheta = 0, \pi$), and reaches its maximum at broadside ($\vartheta = \pi/2$).

Key properties

$D^{(1)} = 1.64$ (2.15 dBi)

lowest directivity of all five configurations

HPBW⁽¹⁾ $\approx 78^\circ$

elevation beamwidth; no azimuth selectivity

No free parameters — pattern cannot be steered or shaped. Azimuthal symmetry means no horizontal discrimination. These limitations motivate all subsequent array configurations.

2.4 Configuration 2 & 3: Linear Dipole Array

A single half-wave dipole radiates equally in all azimuthal directions. Placing N identical half-wave dipoles along $\hat{\mathbf{z}}$ with uniform spacing d and feeding them coherently produces, through wave interference, highly directional pattern. By controlling the progressive phase shift δ between neighbouring elements, the main beam can be steered to any elevation angle ϑ_0 without physically moving the antennas. Config. 2 is the special case $\delta = 0$ (broadside); Config. 3 allows $\delta \neq 0$ (steered).

Physical setup

N half-wave dipoles are placed along $\hat{\mathbf{z}}$ at positions $z_n = nd$, $n = 0, \dots, N-1$, with uniform inter-element spacing d . Every element is fed with the same amplitude I_0 and a progressive phase shift $n\delta$:

$$I_n(t) = I_0 \cos(\omega t + n\delta), \quad n = 0, \dots, N-1.$$

The two configurations studied are:

- **Config. 2** ($\delta = 0$): uniform phase, main lobe broadside.
- **Config. 3** ($\delta \neq 0$): progressive phase shift, main lobe steered to ϑ_0 .

Phase variable

In the far-field limit $r \gg Nd$, all rays to the observation point are parallel at angle ϑ to $\hat{\mathbf{z}}$. Element n is a distance $nd \cos \vartheta$ further from the observation point than element 0, introducing a phase lag $nkd \cos \vartheta$. Combined with the intentional shift $n\delta$, the total phase of element n relative to element 0 is $n\psi$, where the *phase variable* is

$$\psi = kd \cos \vartheta + \delta.$$

All directional properties of the array depend on ϑ and δ only through ψ .

Array Factor: derivation

The total far-field is the single-element field multiplied by the coherent sum over all N elements:

$$\mathbf{E}_{\text{tot}} = \mathbf{E}_1 \sum_{n=0}^{N-1} e^{in\psi} = \mathbf{E}_1 \cdot S.$$

$S = \sum_{n=0}^{N-1} e^{in\psi}$ is a geometric series. Applying $\sum_{n=0}^{N-1} q^n = (1 - q^N)/(1 - q)$ with $q = e^{i\psi}$ and factoring the common phase $e^{i(N-1)\psi/2}$ using $1 - e^{i\theta} = -2i \sin(\theta/2) e^{i\theta/2}$:

$$S = e^{i(N-1)\psi/2} \frac{\sin(N\psi/2)}{\sin(\psi/2)}, \quad |S| = \left| \frac{\sin(N\psi/2)}{\sin(\psi/2)} \right|.$$

$|S|$ reaches its maximum value N when $\psi = 0$ (verified by L'Hôpital's rule). Normalising by N gives the Array Factor:

$$\boxed{\text{AF}(\vartheta; \delta) = \frac{\sin(N\psi/2)}{N \sin(\psi/2)}, \quad \psi = kd \cos \vartheta + \delta.} \quad (2)$$

Steering condition

The main lobe is at $\psi = 0$. To place it at a desired direction ϑ_0 , we impose $\psi = 0$ at $\vartheta = \vartheta_0$:

$$kd \cos \vartheta_0 + \delta = 0 \quad \implies \quad \boxed{\delta = -kd \cos \vartheta_0.}$$

Substituting into ψ gives $\psi = kd(\cos \vartheta - \cos \vartheta_0)$, so the Array Factor becomes

$$\text{AF}(\vartheta; \vartheta_0) = \frac{\sin\left(\frac{Nkd}{2}(\cos \vartheta - \cos \vartheta_0)\right)}{N \sin\left(\frac{kd}{2}(\cos \vartheta - \cos \vartheta_0)\right)}.$$

Config. 2 is recovered by setting $\vartheta_0 = \pi/2$ (broadside), which gives $\delta = 0$ and $\psi = kd \cos \vartheta$. No antenna moves; only the electronic phase δ changes.

Total pattern and gain

Since $|\mathbf{E}_{\text{tot}}|^2 \propto |\mathbf{E}_1|^2 \cdot |S|^2$, the pattern multiplication rule gives the normalised total pattern:

$$\tilde{U}^{(k)}(\vartheta; \vartheta_0) = g(\vartheta) \cdot |\text{AF}(\vartheta; \vartheta_0)|^2, \quad (3)$$

where $k = 2$ for $\vartheta_0 = \pi/2$ and $k = 3$ otherwise. The element factor $g(\vartheta)$ and the Array Factor are completely independent: N, d, δ affect only AF; the dipole physics affect only g . The power gain is

$$G^{(k)}(\vartheta; \vartheta_0) = D^{(k)}(\vartheta_0) \cdot g(\vartheta) \cdot |\text{AF}(\vartheta; \vartheta_0)|^2,$$

where the directivity $D^{(k)}(\vartheta_0)$ is given in the key properties below.

Key properties

Main lobe and nulls. $\text{AF} = 1$ at $\psi = 0$. Nulls at $\cos \vartheta_{\text{null}} = \cos \vartheta_0 \pm m\lambda/Nd$, $m = 1, 2, \dots$

- **Config. 2** ($\delta = 0$): main lobe broadside, $\vartheta_0 = \pi/2$.
- **Config. 3** ($\delta = -kd \cos \vartheta_0$): main lobe steered to ϑ_0 .

Half-power beamwidth. For large N , $\text{AF} \approx \text{sinc}(N\psi/2)$. Setting $|\text{AF}| = 1/\sqrt{2}$ gives $N\psi_{\text{HP}}/2 = 1.391$, hence via $\Delta\psi = kd \sin \vartheta_0 \Delta\vartheta$:

$$\text{HPBW}(\vartheta_0) \approx \frac{0.886 \lambda}{Nd \sin \vartheta_0}. \quad (4)$$

- **Config. 2:** $\text{HPBW}^{(2)} \approx 0.886 \lambda / Nd$.
- **Config. 3:** $\text{HPBW}^{(3)} \approx 0.886 \lambda / (Nd \sin \vartheta_0) \geq \text{HPBW}^{(2)}$. Beam broadens away from broadside due to aperture foreshortening.

First sidelobe level. The first sidelobe occurs at the first maximum of $|\text{AF}|$ after the main lobe. For large N , $\sin(\psi/2) \approx \psi/2$ and the Array Factor approximates a sinc function:

$$\text{AF}(\psi) \approx \frac{\sin(N\psi/2)}{N\psi/2}.$$

The sidelobe maxima are found by differentiating and setting to zero:

$$\frac{d}{d\psi} \left[\frac{\sin(N\psi/2)}{N\psi/2} \right] = \frac{(N\psi/2) \cos(N\psi/2) - \sin(N\psi/2)}{(N\psi/2)^2} = 0,$$

which requires:

$$\tan\left(\frac{N\psi}{2}\right) = \frac{N\psi}{2}.$$

The first non-trivial solution occurs near $N\psi_{\text{SL}}/2 \approx 3\pi/2$, where $\tan(3\pi/2) \rightarrow \infty$. Evaluating the Array Factor at this point:

$$|\text{AF}|_{\text{SL}} = \left| \frac{\sin(3\pi/2)}{3\pi/2} \right| = \frac{1}{3\pi/2} = \frac{2}{3\pi}.$$

The first sidelobe level relative to the main lobe ($|\text{AF}|_{\text{max}} = 1$) is therefore:

$$\text{SLL} = 20 \log_{10} \left(\frac{2}{3\pi} \right) \approx -13.3 \text{ dB}. \quad (5)$$

Identical for Configs. 2 and 3; universal property of uniform arrays, independent of N, d, δ .

Directivity ($d = \lambda/2$, large N).

- **Config. 2:** $D^{(2)} \approx N$.
- **Config. 3:**

$$: D^{(3)}(\vartheta_0) \approx N \sin \vartheta_0 \leq D^{(2)} \quad (6)$$

Steering reduces directivity by $\sin \vartheta_0$.

Grating lobes. Spurious maxima at

$$\cos \vartheta_g = \cos \vartheta_0 \pm m\lambda/d \quad (7)$$

Identical for Configs. 2 and 3. For $d = \lambda/2$: nearest grating lobe requires $|\cos \vartheta_g| > 1$; unphysical for all ϑ_0 . Standard choice $d = \lambda/2$ guarantees grating-lobe-free operation.

2.5 Configurations 4 & 5: Planar Dipole Array

A linear array controls the beam only in the elevation plane (ϑ). To achieve independent control of both elevation *and* azimuth (φ), we extend to a two-dimensional rectangular array of $N_x \times N_z$ half-wave dipoles. Config. 4 is the special case $\boldsymbol{\delta} = \mathbf{0}$ (no steering); Config. 5 allows $\boldsymbol{\delta} \neq \mathbf{0}$ (full 2D steering).

Physical setup

Place N_x columns along $\hat{\mathbf{x}}$ with spacing d_x and N_z rows along $\hat{\mathbf{z}}$ with spacing d_z . Element (m, n) sits at position $\mathbf{r}_{mn} = md_x\hat{\mathbf{x}} + nd_z\hat{\mathbf{z}}$ and is fed with amplitude I_0 and progressive phase $m\delta_x + n\delta_z$:

$$I_{mn}(t) = I_0 \cos(\omega t + m\delta_x + n\delta_z), \quad m = 0, \dots, N_x - 1, \quad n = 0, \dots, N_z - 1.$$

The two configurations are:

- **Config. 4** ($\delta_x = \delta_z = 0$): main lobe broadside.
- **Config. 5** ($\delta_x, \delta_z \neq 0$): main lobe steered to (ϑ_0, φ_0) .

Phase variables

In the far-field limit $r \gg N_x d_x, N_z d_z$, all rays are approximately parallel. The path-length difference from element (m, n) to the observation point (ϑ, φ) is $md_x \sin \vartheta \cos \varphi + nd_z \cos \vartheta$. The total phase of element (m, n) relative to element $(0, 0)$ is $m\psi_x + n\psi_z$, where the two phase variables are:

$$\psi_x = kd_x \sin \vartheta \cos \varphi + \delta_x, \quad \psi_z = kd_z \cos \vartheta + \delta_z. \quad (8)$$

ψ_x governs the azimuthal interference (along $\hat{\mathbf{x}}$); ψ_z governs the elevation interference (along $\hat{\mathbf{z}}$). They are completely independent.

Array Factor: derivation

The total far-field is the single-element field multiplied by the coherent double sum:

$$\mathbf{E}_{\text{tot}} = \mathbf{E}_1 \underbrace{\left(\sum_{m=0}^{N_x-1} e^{im\psi_x} \right)}_{S_x} \underbrace{\left(\sum_{n=0}^{N_z-1} e^{in\psi_z} \right)}_{S_z}.$$

The double sum factorises exactly into two independent geometric series, this is the separability approximation, valid when mutual coupling between elements is neglected. Each series is evaluated identically to the linear array case, giving:

$$|S_x| = \left| \frac{\sin(N_x \psi_x / 2)}{\sin(\psi_x / 2)} \right|, \quad |S_z| = \left| \frac{\sin(N_z \psi_z / 2)}{\sin(\psi_z / 2)} \right|.$$

Normalising each by its maximum value (N_x and N_z respectively) gives the factorised Array Factor:

$$\boxed{\text{AF}(\vartheta, \varphi; \boldsymbol{\delta}) = \text{AF}_x(\psi_x) \cdot \text{AF}_z(\psi_z),} \quad (9)$$

$$\text{AF}_x(\psi_x) = \frac{\sin(N_x \psi_x / 2)}{N_x \sin(\psi_x / 2)}, \quad \text{AF}_z(\psi_z) = \frac{\sin(N_z \psi_z / 2)}{N_z \sin(\psi_z / 2)}.$$

The two factors are completely independent: N_x, d_x, δ_x affect only AF_x ; N_z, d_z, δ_z affect only AF_z .

Steering condition

The main lobe is at $\psi_x = \psi_z = 0$ simultaneously. Imposing this at the desired direction (ϑ_0, φ_0) :

$$\boxed{\delta_x = -kd_x \sin \vartheta_0 \cos \varphi_0, \quad \delta_z = -kd_z \cos \vartheta_0.} \quad (10)$$

Config. 4 is recovered by setting $\delta_x = \delta_z = 0$, which gives the main lobe at $\vartheta_0 = \pi/2$ (broadside) for any φ_0 . The two axes steer independently: δ_z controls elevation and δ_x controls azimuth.

Total pattern and gain

By the pattern multiplication rule:

$$\tilde{U}^{(k)}(\vartheta, \varphi; \boldsymbol{\delta}) = g(\vartheta) \cdot |\text{AF}_x(\psi_x)|^2 \cdot |\text{AF}_z(\psi_z)|^2, \quad (11)$$

where $k = 4$ for $\boldsymbol{\delta} = \mathbf{0}$ and $k = 5$ otherwise. The power gain is:

$$G^{(k)}(\vartheta, \varphi; \boldsymbol{\delta}) = D^{(k)}(\vartheta_0) \cdot g(\vartheta) \cdot |\text{AF}_x(\psi_x)|^2 \cdot |\text{AF}_z(\psi_z)|^2.$$

Key properties

Half-power beamwidth. Each axis behaves independently as a linear array. For large N_x, N_z :

$$\text{HPBW}_x(\vartheta_0, \varphi_0) \approx \frac{0.886 \lambda}{N_x d_x \cos \varphi_0 \sin \vartheta_0}, \quad \text{HPBW}_z(\vartheta_0) \approx \frac{0.886 \lambda}{N_z d_z \sin \vartheta_0}.$$

- **Config. 4:** both beamwidths at their minimum (broadside, $\sin \vartheta_0 = 1$).
- **Config. 5:** both broaden as the beam steers away from broadside.

First sidelobe level. SLL ≈ -13.3 dB along each axis independently. Identical for Configs. 4 and 5; universal property of uniform weighting.

Directivity ($d_x = d_z = \lambda/2$, large N).

- **Config. 4:**

$$D^{(4)} \approx \pi N_x N_z / 2 \quad (12)$$

- **Config. 5:**

$$: D^{(5)}(\vartheta_0) \approx \pi N_x N_z \sin \vartheta_0 / 2 \leq D^{(4)} \quad (13)$$

Grating lobes. Spurious maxima appear when $\psi_x = 2m\pi$ or $\psi_z = 2m\pi$ simultaneously. Their directions along each axis satisfy:

$$\sin \vartheta_g \cos \varphi_g = \sin \vartheta_0 \cos \varphi_0 \pm \frac{m\lambda}{d_x}, \quad \cos \vartheta_g = \cos \vartheta_0 \pm \frac{m\lambda}{d_z}.$$

- **Config. 4:** grating lobes symmetric about broadside in both planes.
- **Config. 5:** grating lobe positions shift with (ϑ_0, φ_0) .

The choice $d_x = d_z = \lambda/2$ guarantees grating-lobe-free operation in both planes for all steering angles.

2.6 Directivity and Gain: Summary

Having derived the gain function $G^{(k)}$ for each configuration, we now compare them directly. Table 2 summarises the key figures of merit for all five configurations at $d = \lambda/2$.

Table 2: Comparison of the five configurations at $d = \lambda/2$, broadside, large N .

Config.	Description	$D^{(k)}$	HPBW	Steering
1	Single dipole	1.64	78°	none
2	Linear, $\delta = 0$	N	$0.886\lambda/Nd$	none
3	Linear, $\delta \neq 0$	$N \sin \vartheta_0$	$0.886\lambda/(Nd \sin \vartheta_0)$	elevation only
4	Planar, $\delta = \mathbf{0}$	$\pi N_x N_z / 2$	$0.886\lambda/N_x d_x, 0.886\lambda/N_z d_z$	none
5	Planar, $\delta \neq \mathbf{0}$	$\pi N_x N_z \sin \vartheta_0 / 2$	broader than Config. 4	elevation + azimuth

Key observations. Directivity grows with aperture. Every additional element increases directivity linearly: $D^{(2)} \approx N$ for the linear array and $D^{(4)} \approx \pi N_x N_z / 2$ for the planar array. A planar array of $N_x \times N_z$ elements achieves a directivity $\pi N_x N_z / 2$ times higher than a linear array of N_x elements alone — the second dimension multiplies the directivity.

Steering always reduces directivity. Configs. 3 and 5 pay a directivity penalty of $\sin \vartheta_0 \leq 1$ when steered away from broadside. This is a fundamental geometric consequence of aperture foreshortening, not a limitation of the design. Maximum directivity is always achieved at broadside.

The planar array is the most capable configuration. Config. 5 is the only configuration that achieves simultaneous control of both elevation and azimuth, and it has the highest directivity of all five. The price paid is complexity: $2N$ phase shifters are required instead of N .

Config. 1 is the reference. The single dipole has no directional selectivity in the horizontal plane. Its gain $G^{(1)} = 1.64 g(\vartheta)$ appears as a factor in every other configuration through the element factor, making it the true building block of the entire analysis.

2.7 Campus Coverage Optimisation

Using the gain functions $G^{(k)}$ derived, we now pose the campus coverage optimisation problem precisely. The goal is to determine both the optimal array configuration k^* and, for steerable configurations, the optimal phase shifts δ^* .

Physical model of the campus

Let $\Omega \subset \mathbb{R}^2$ denote the campus footprint and $\Omega_B \subset \Omega$ the union of building regions. We represent each of the M buildings by a reference point (x_i, y_i) , $i = 1, \dots, M$, extracted from the campus map. The antenna is fixed at height h above position (x_0, y_0) . The geometry to building i is:

$$r_i = \sqrt{\rho_i^2 + h^2}, \quad \vartheta_i = \arccos \frac{h}{r_i}, \quad \varphi_i = \text{atan2}(y_i - y_0, x_i - x_0),$$

where ρ_i is the horizontal distance. The received power flux density at building i for configuration k follows from the free-space link budget; the total power P_{in} spread over a sphere of radius r_i , concentrated by the antenna gain $G^{(k)}$:

$$S_i^{(k)}(\delta) = \frac{G^{(k)}(\vartheta_i, \varphi_i; \delta) P_{\text{in}}}{4\pi r_i^2} \quad [\text{W m}^{-2}].$$

Building i is *covered* when $S_i^{(k)} \geq P_{\text{min}}$, where $P_{\text{min}} [\text{W m}^{-2}]$ is the minimum power flux density required for a stable connection.

Objective function

Coverage score. The ideal objective counts covered buildings via the Heaviside function H :

$$C_{\text{ideal}}^{(k)}(\delta) = \sum_{i=1}^M H(S_i^{(k)}(\delta) - P_{\text{min}}).$$

This is not differentiable: $\partial H / \partial \delta_n = 0$ almost everywhere, so gradient-based optimisation stalls immediately. We replace H with the smooth sigmoid $\sigma_\alpha(x) = 1/(1 + e^{-\alpha x})$, which satisfies $\lim_{\alpha \rightarrow \infty} \sigma_\alpha = H$ and

has non-zero derivative $d\sigma_\alpha/dx = \alpha\sigma_\alpha(1 - \sigma_\alpha) > 0$ everywhere. Defining the normalised coverage ratio $\xi_i^{(k)} = S_i^{(k)}/P_{\min}$, the smooth coverage score is:

$$Q^{(k)}(\boldsymbol{\delta}) = \frac{1}{M} \sum_{i=1}^M \sigma_\alpha(\xi_i^{(k)}(\boldsymbol{\delta}) - 1) \approx \frac{\text{number of covered buildings}}{M}.$$

Efficiency score. Coverage alone is not sufficient, an antenna that radiates strongly everywhere scores well on $Q^{(k)}$ but wastes power on roads, fields, and open areas. We therefore define the fraction of total radiated power delivered to building regions:

$$\eta^{(k)}(\boldsymbol{\delta}) = \frac{\int_{\Omega_B} S^{(k)}(x, y; \boldsymbol{\delta}) dx dy}{\int_{\Omega} S^{(k)}(x, y; \boldsymbol{\delta}) dx dy}.$$

A configuration that concentrates its beam on buildings has $\eta^{(k)} \rightarrow 1$; one that sprays power everywhere has $\eta^{(k)} \rightarrow |\Omega_B|/|\Omega| \ll 1$.

Combined objective. We combine coverage and efficiency into a single scalar objective with non-negative weights $w_c + w_\eta = 1$:

$$J_k(\boldsymbol{\delta}) = w_c Q^{(k)}(\boldsymbol{\delta}) + w_\eta \eta^{(k)}(\boldsymbol{\delta}).$$

The choice $w_c \gg w_\eta$ prioritises coverage; $w_\eta \gg w_c$ prioritises beam efficiency. In practice both are set to 1/2 as a neutral baseline.

Solution procedure

Step 1; optimise phase shifts for steerable configurations. Configurations 3 and 5 have free phase parameters. For each, find the optimal phase shifts by gradient ascent on J_k :

$$\boldsymbol{\delta}_k^* = \arg \max_{\boldsymbol{\delta} \in [-\pi, \pi]^N} J_k(\boldsymbol{\delta}), \quad k \in \{3, 5\}.$$

The gradient ascent update rule is:

$$\boldsymbol{\delta}^{(j+1)} = \boldsymbol{\delta}^{(j)} + \mu \nabla_{\boldsymbol{\delta}} J_k(\boldsymbol{\delta}^{(j)}),$$

with step size $\mu > 0$ and initialisation $\boldsymbol{\delta}^{(0)} = \mathbf{0}$. The gradient of $Q^{(k)}$ with respect to δ_n is:

$$\frac{\partial Q^{(k)}}{\partial \delta_n} = \frac{1}{M} \sum_{i=1}^M \underbrace{\alpha^2 \sigma_i (1 - \sigma_i)}_{\text{bell-shaped: largest near threshold}} \cdot \frac{P_{\text{in}}}{4\pi r_i^2 P_{\min}} \cdot \frac{\partial G_i^{(k)}}{\partial \delta_n},$$

which is non-zero everywhere and largest for buildings near the coverage margin $\xi_i \approx 1$ — exactly where the optimiser should focus. To avoid the near-flat landscape at large α , we use *homotopy continuation*: start with $\alpha_0 = 3\text{--}5$, converge, then set $\alpha_{j+1} = 2\alpha_j$ and re-optimize from the previous solution, repeating until $\alpha_j > 50$. Configs. 1, 2, 4 have no free parameters: $\boldsymbol{\delta}_1^* = \boldsymbol{\delta}_2^* = \boldsymbol{\delta}_4^* = \mathbf{0}$.

Step 2 — select the best configuration. Evaluate $J_k(\boldsymbol{\delta}_k^*)$ for all five configurations and select the winner:

$$k^* = \arg \max_{k \in \{1, 2, 3, 4, 5\}} J_k(\boldsymbol{\delta}_k^*).$$

The result $\boldsymbol{\delta}_{k^*}^*$ is the optimal phase configuration: the electronic beam-steering pattern that maximises the combined coverage and efficiency objective for the best array architecture.

3 Simulation Method

All simulations were implemented in Python 3.12 using NumPy for numerical computation and Matplotlib for visualisation.

Preliminary Antenna Configuration Simulations

Before applying the antenna model to the campus environment, a set of preliminary simulations was conducted to characterise the radiation behaviour of different array configurations in isolation. These simulations served to validate the array factor implementation and to build physical intuition about how array geometry influences beam shape and directivity. These are included in the Jupyter Notebook called preliminary simulations, whereas the rest of the code is included in the Jupyter Notebook called campus simulations.

Campus geometry

The campus was modelled as a set of $M = 60$ buildings, each represented by its centre point (x_i, y_i) in metres relative to the antenna position. Building coordinates were extracted from GPS measurements using the equirectangular approximation:

$$x = (\lambda_{\text{lon}} - \lambda_{\text{lon},0}) \cos(\phi_{\text{lat},0}) \cdot 111320 \text{ m}, \quad y = (\phi_{\text{lat}} - \phi_{\text{lat},0}) \cdot 111320 \text{ m},$$

where $(\phi_{\text{lat},0}, \lambda_{\text{lon},0})$ is the antenna GPS position, taken as the origin. The antenna was placed at height $h = 10$ m above the origin. For each building i , the 3D geometry $(\vartheta_i, \varphi_i, r_i)$ was computed from (x_i, y_i, h) , and the received power flux density evaluated as:

$$S_i^{(k)} = \frac{G^{(k)}(\vartheta_i, \varphi_i; \boldsymbol{\delta}) P_{\text{in}}}{4\pi r_i^2},$$

with $P_{\text{in}} = 0.1$ W (100 mW, consistent with ETSI EN 300 328 for 2.4 GHz small-cell transmitters), operating frequency $f = 2.4$ GHz ($\lambda \approx 0.125$ m), and inter-element spacing $d = \lambda/2$ throughout. The minimum required power flux density was set to $P_{\text{min}} = 2.5 \times 10^{-7}$ W m⁻², the threshold at which steerable configurations first outperform the single dipole on the combined objective.

Objective function

The smooth coverage score $Q^{(k)}$ was computed using the sigmoid approximation $\sigma_\alpha(x) = (1 + e^{-\alpha x})^{-1}$ with sharpness $\alpha = 10$. The combined objective was $J_k = w_c Q^{(k)} + w_\eta \eta^{(k)}$ with $w_c = w_\eta = 0.5$.

Phase optimisation

For Configs. 3 and 5, the phase parameters were optimised by gradient ascent on J_k . Gradients were approximated by central finite differences with step size $\varepsilon = 10^{-5}$ rad. The update rule was:

$$\boldsymbol{\delta}^{(j+1)} = \text{clip}\left(\boldsymbol{\delta}^{(j)} + \mu \nabla_{\boldsymbol{\delta}} J_k, -\pi, \pi\right),$$

with step size $\mu = 0.1$ and initialisation $\boldsymbol{\delta}^{(0)} = \mathbf{0}$. Homotopy continuation was applied: α was initialised at 3 and doubled every 300 iterations until $\alpha > 50$, for a total of 1500 gradient steps. Config. 3 was optimised over a single scalar δ ; Config. 5 over two scalars (δ_x, δ_z) .

Spatial coverage maps

The radiation maps of Figure 9 were generated by evaluating $S_i^{(k)}$ on a 160×160 uniform grid over the campus domain $x \in [-750, 350]$ m, $y \in [-450, 570]$ m, with buildings overlaid as coloured rectangles indicating coverage status.

4 Results and Discussion

4.1 Linear Array with Uniform Spacing, Broadside Configuration

We begin with Config. 2: a uniform linear array (ULA) of N half-wave dipole elements arranged along the z -axis, with equal inter-element spacing d and zero phase offset ($\delta = 0$), so that the main radiation lobe is directed perpendicular to the array axis (broadside).

Effect of Array Size

Figure 1 shows the simulated polar radiation patterns for ULAs of $N = 4, 8$, and 16 elements at fixed spacing $d = \lambda/2$. The patterns are in qualitative agreement with the theoretical predictions of Section 2.4.

The main lobe narrows with increasing N , consistent with the analytical expression for the half-power beamwidth at broadside ($\vartheta_0 = \pi/2$, $d = \lambda/2$):

$$\text{HPBW} \approx \frac{1.772}{N} \text{ rad} \approx \frac{101.5^\circ}{N}. \quad (14)$$

The sidelobe structure becomes more complex as N increases, though the first sidelobe level remains at the theoretical value of -13.3 dB characteristic of uniform amplitude weighting, independent of N . This constitutes the fundamental trade-off of uniform arrays: narrowing the main beam comes at the cost of a more pronounced sidelobe pattern.

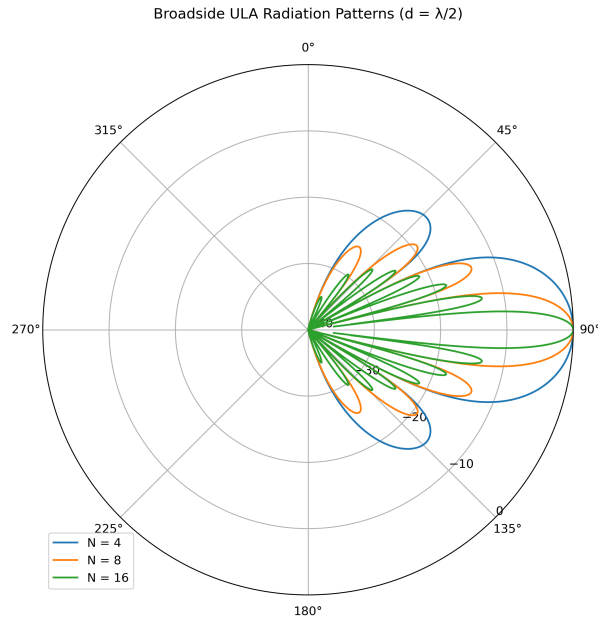


Fig. 1: Polar radiation patterns (normalized far-field intensity, dB scale) for broadside ULAs of $N = 4, 8$, and 16 half-wave dipole elements at inter-element spacing $d = \lambda/2$. The main lobe narrows and the sidelobe structure becomes more complex with increasing N . All patterns are plotted in the H-plane ($\phi = 90^\circ$).

Effect of Inter-Element Spacing

Figure 2 illustrates the effect of varying d at fixed $N = 8$. For $d = \lambda/4$, the effective aperture is reduced and the main lobe broadens significantly. For $d = \lambda/2$, the optimal trade-off between beamwidth and sidelobe suppression is achieved. For $d = \lambda$, grating lobes appear as additional main lobes of equal amplitude at angles ϑ_g satisfying $\cos \vartheta_g = \cos \vartheta_0 \pm \frac{m\lambda}{d}$, $m = 1, 2, 3, \dots$. For broadside steering, this predicts grating lobes at $\vartheta_g = 0^\circ$ and 180° , which are clearly visible in the right panel of Figure 2. These represent a serious degradation in performance, confirming that $d = \lambda/2$ is the standard design choice.

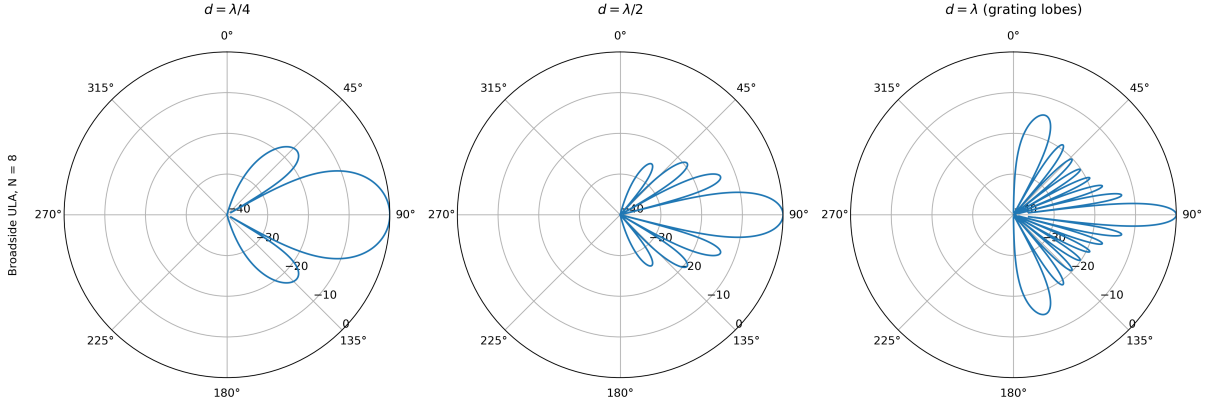


Fig. 2: Effect of inter-element spacing on the broadside ULA radiation pattern ($N = 8$). Left: $d = \lambda/4$ (broadened main lobe). Centre: $d = \lambda/2$ (optimal). Right: $d = \lambda$ (grating lobes appear at $\vartheta \approx 0^\circ$ and 180° , indicated by arrows).

Planar Array Configuration

Figure 3 compares three configurations: 4×4 , 4×8 , and 4×16 . In all cases, the number of elements along the x -direction is held constant while the number along the y -direction is increased.

The results clearly demonstrate that the dipole distribution along each axis independently controls the radiation pattern in its corresponding direction. Increasing N_y progressively narrows the beam in the azimuthal regions governed by the y -axis, while the contribution from the x -axis remains unchanged. Consequently, the overall pattern becomes increasingly directive in ϕ , with sharper main lobes and more pronounced sidelobes as N_y increases.

This behaviour is a direct consequence of the separable structure of the array factor,

$$AF(\theta, \phi) = AF_x(\theta, \phi) \cdot AF_y(\theta, \phi),$$

where each term acts as an independent linear array. The x -directed array shapes the variation associated with $\cos \phi$, while the y -directed array shapes the variation associated with $\sin \phi$. The azimuthal radiation pattern therefore emerges from the product of two orthogonal contributions, each controlled solely by the element distribution along its respective axis.

As a result, planar arrays provide independent control over beam shaping in orthogonal directions: modifying N_x or N_y selectively alters the radiation pattern without coupling between the two axes.

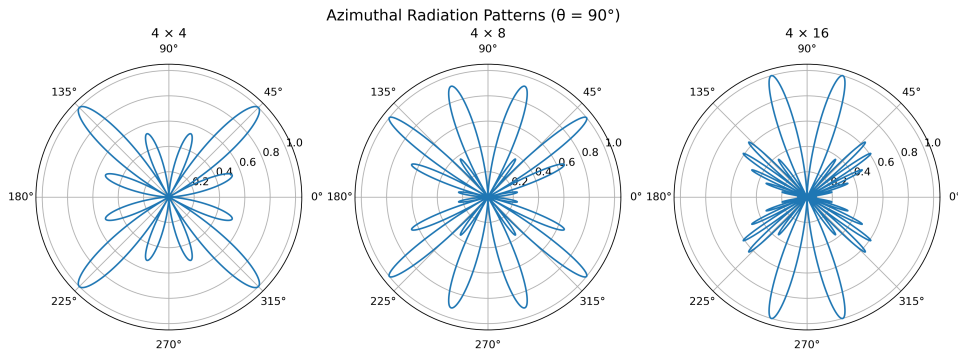


Fig. 3: Normalised azimuthal radiation patterns ($\theta = 90^\circ$) for planar arrays with $d_x = d_y = \lambda/2$. The three panels correspond to (left) 4×4 , (centre) 4×8 , and (right) 4×16 configurations. Increasing the number of elements along the y -axis progressively narrows the beam in the corresponding azimuthal direction, while the x -axis contribution remains unchanged.

4.2 Electronic Beam Steering via Phase Shaping

Configs. 3 and 5 introduce electronic beam steering by applying a progressive inter-element phase shift δ , steering the main lobe to a desired direction without mechanical movement. Physically, this phase progression imposes a controlled phase gradient across the array aperture, such that radiation from each element adds constructively in a chosen direction and destructively elsewhere. The beam direction is therefore determined by the condition that the phase delay due to element spacing is exactly cancelled by the applied phase shift.

Linear Array Steering

Figure 4 shows the radiation patterns of a 16-element ULA ($d = \lambda/2$) steered to $\vartheta_0 = 90^\circ$ (broadside), 60° , 30° , and 0° (endfire). The main lobe successfully tracks the target angle in all cases, confirming that the imposed phase shift $\delta = -kd \cos \vartheta_0$ effectively tilts the direction of constructive interference. In other words, the array is “pre-delayed” so that waves travelling in the desired direction arrive in phase.

As the beam moves away from broadside, the beamwidth broadens, consistent with the analytical prediction:

$$\text{HPBW}(\vartheta_0) \approx \frac{0.886 \lambda}{Nd \sin \vartheta_0},$$

which diverges as $\vartheta_0 \rightarrow 0^\circ$. This broadening, known as *beam squint* or *scan loss*, reflects the reduction in the effective projected aperture as the beam approaches endfire. Near endfire, pattern asymmetries also appear because the visible range of ψ is clipped to $[-\pi, \pi]$, limiting the range of constructive interference and reducing the effective aperture.

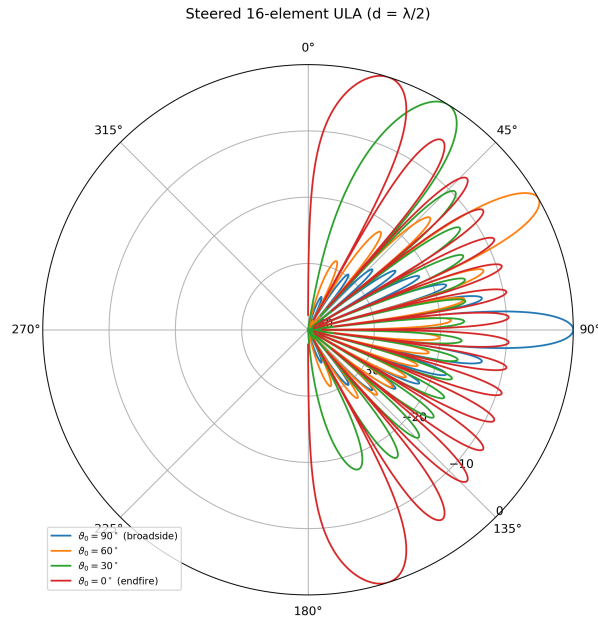


Fig. 4: Radiation patterns of a 16-element ULA ($d = \lambda/2$) steered electronically to $\vartheta_0 = 90^\circ$ (broadside, $\delta = 0$), 60° , 30° , and 0° (endfire) by applying progressive phase shifts $\delta = -kd \cos \vartheta_0$. Patterns shown in polar form (normalised dB). Note the progressive beamwidth broadening as the beam approaches endfire.

Planar Array Steering

Figure 5 extends this to a 2D steered 8×8 planar array directed to $(\vartheta_0 = 30^\circ, \varphi_0 = 45^\circ)$. Steering is now achieved by applying independent phase gradients along each axis, allowing the beam to be directed simultaneously in elevation and azimuth. Specifically, the phase shifts δ_x and δ_z compensate for the path differences associated with propagation in the target direction, ensuring constructive interference at (ϑ_0, φ_0) .

The main lobe correctly points to the target direction, confirming that independent control of δ_x and δ_z provides full 2D steering capability. Beam squint now broadens the beam footprint in two dimensions rather than one, consistent with the $1/\sin \vartheta_0$ scaling of both HPBW_x and HPBW_z . This reflects the fact that the effective aperture is reduced along both axes when steering away from broadside.

Steered 8×8 Planar Array ($\theta_0=30^\circ$, $\varphi_0=45^\circ$)



Fig. 5: 3D far-field pattern of an 8×8 planar array steered to $(\vartheta_0 = 30^\circ, \varphi_0 = 45^\circ)$ by applying independent phase gradients $\delta_x = -kd_x \sin \vartheta_0 \cos \varphi_0$ and $\delta_z = -kd_z \cos \vartheta_0$. The main lobe points to the target direction while preserving the array gain.

4.3 Directivity Analysis and Comparison with a Single Dipole

Figure 6 shows the simulated directivity as a function of array size N for broadside ULAs with $d = \lambda/2$, alongside the analytical approximation $D \approx N$ and the single half-wave dipole reference of 2.15 dBi. The simulated values follow the analytical scaling closely across all tested array sizes, confirming that directivity increases approximately linearly with the number of radiating elements.

Physically, this trend arises because increasing N enlarges the effective aperture of the array. A larger aperture concentrates the same total radiated power into a narrower angular region, reducing the beamwidth and therefore increasing the peak radiation intensity. As a result, each doubling of the number of elements produces an approximate 3 dB increase in directivity, consistent with aperture theory.

Compared to the single half-wave dipole, even modest array sizes significantly enhance performance. For example, a 16-element array already provides substantially higher directivity than the dipole baseline, demonstrating the strong advantage of spatially distributed radiation over a single-element radiator.

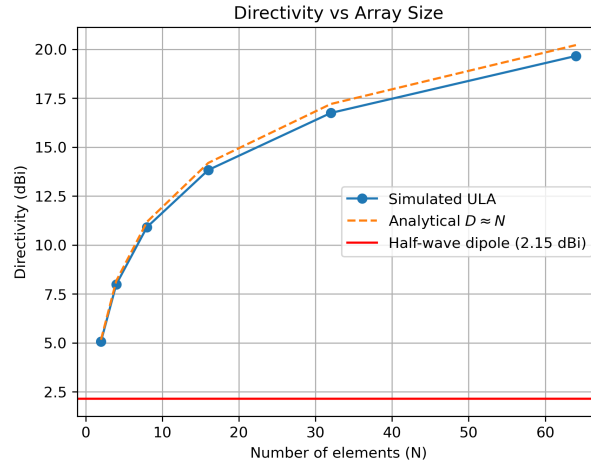


Fig. 6: Simulated directivity (dBi) as a function of array size N for broadside ULAs with $d = \lambda/2$ (blue circles). The dashed line shows the analytical approximation $D \approx N$ converted to dBi. The horizontal red line marks the single half-wave dipole reference (2.15 dBi).

Figure 7 shows the directivity of the 8×8 planar array as a function of steering angle ϑ_0 in the E-plane, normalised to the broadside value. The directivity decreases as the beam is steered away from broadside because the effective projected aperture reduces with angle. In other words, when the beam is tilted, fewer elements contribute constructively in the intended direction, reducing the coherent gain of the array.

The results follow the expected $\sin \vartheta_0$ dependence, with directivity remaining close to its maximum within approximately $\pm 60^\circ$ from broadside. Within this region, the array maintains efficient constructive interference across most of its aperture. Beyond this range, scan loss becomes significant: the phase alignment between elements becomes less effective, leading to reduced peak gain and a widening of the main lobe.

The -3 dB threshold highlights the practical steering limit of the array, beyond which performance degrades rapidly due to the combination of reduced effective aperture and increased sidelobe contribution. This demonstrates that while planar arrays offer full electronic steering capability, their usable angular range is fundamentally constrained by geometry and phase coherence.

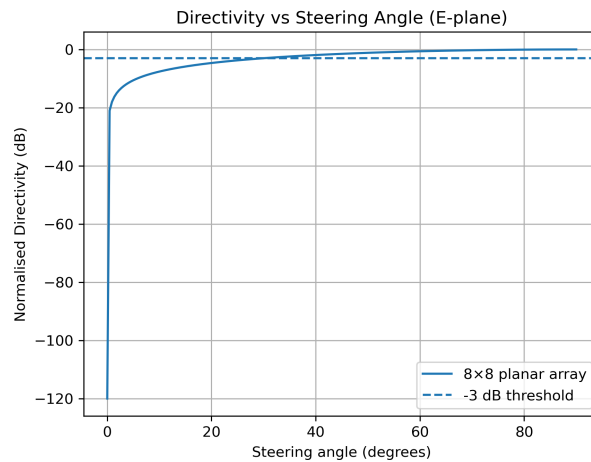


Fig. 7: Simulated directivity $D(\vartheta_0)$ of an 8×8 planar array ($d_x = d_z = \lambda/2$) as a function of steering angle ϑ_0 in the E-plane ($\varphi_0 = 0^\circ$), normalised to the broadside value D_0 . The dashed line indicates the -3 dB threshold, corresponding to the conventional $\pm 60^\circ$ operational envelope of phased arrays.

4.4 Campus Coverage Optimisation

This section presents the results of the campus coverage optimization. For each of the five configurations, we evaluate the combined objective $J_k(\delta_k^*)$ and identify the optimal array architecture and phase configuration.

Campus Geometry

The campus contains $M = 60$ buildings, with the antenna placed at height $h = 10\text{m}$ above the origin. The minimum required power flux density is $P_{\min} = 2.5 \cdot 10^{-7} \text{ W m}^{-2}$, as this value represents the typical minimal power for receivers. Below is a map of these buildings on the campus with the antenna placed at the origin.

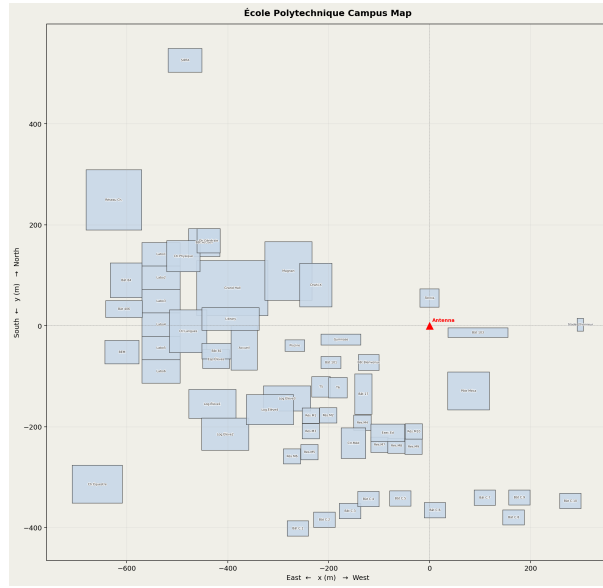


Fig. 8: Campus map showing the $M = 60$ buildings and the antenna at the origin.

Coverage Results

Table 3 summarises the coverage score $Q^{(k)}$, combined objective J_k , and how many buildings covered for all five configurations at their optimal phase settings.

Table 3: Coverage score $Q^{(k)}$, combined objective J_k , and number of buildings covered for all five configurations at optimal phase settings ($w_c = w_\eta = 0.5$).

Config.	Description	$Q^{(k)}$	J_k	Covered Buildings
1	Single dipole	0.1758	0.5879	10/60
2	Linear, $\delta = 0$	0.6839	0.8419	40/60
3	Linear, steered	0.6960	0.8480	41/60
4	Planar, $\delta = \mathbf{0}$	0.1543	0.5771	9/60
5	Planar, steered	0.7213	0.8607	44/60

The optimal configuration is Config. 5, achieving a combined objective of $J_{k^*} = 0.8607$, with 73% of buildings covered and 72.13% of radiated power directed towards building regions.

Spatial Coverage Map

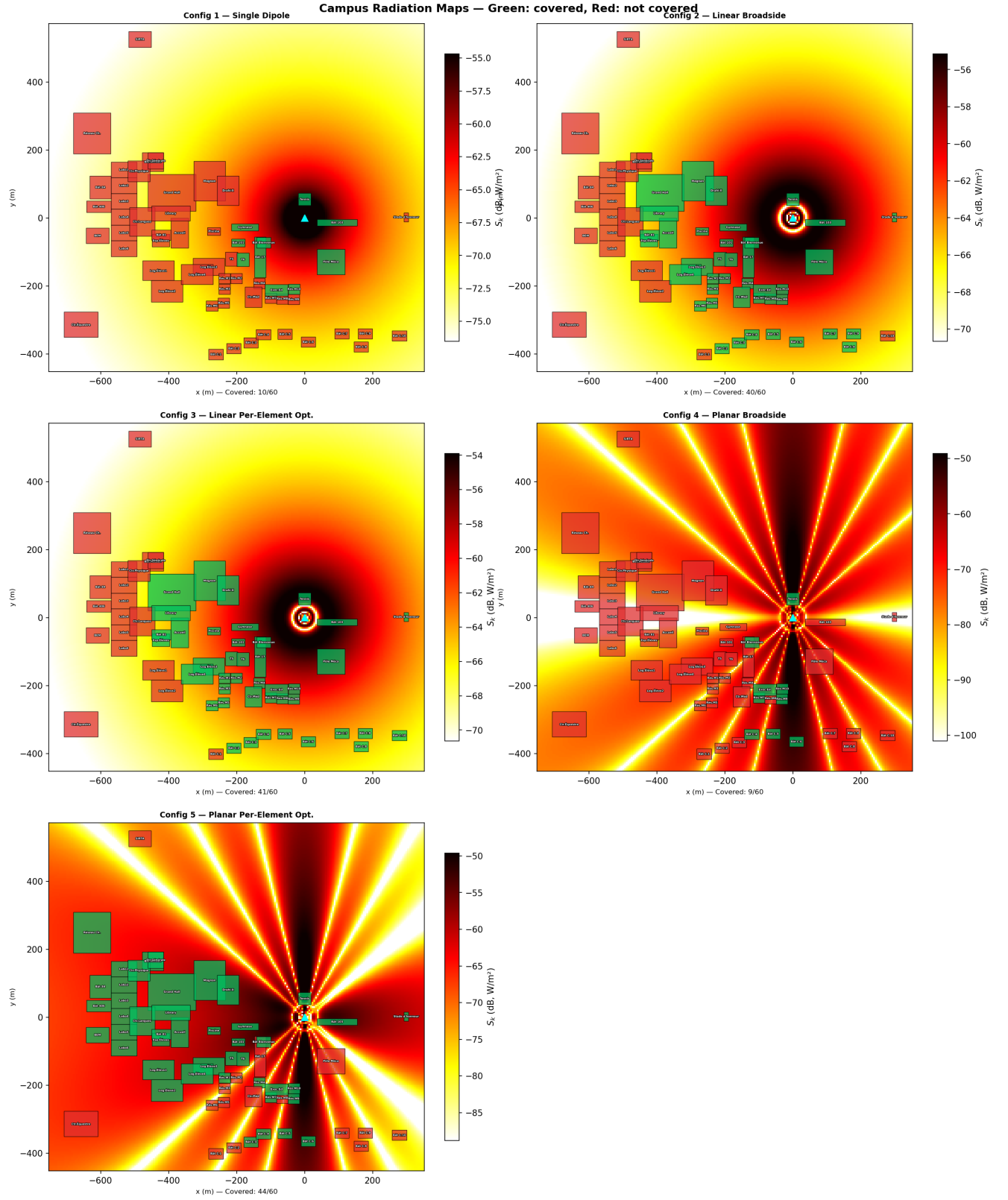


Fig. 9: Spatial map of received power flux density $S^{(k^*)}(x, y; \delta_{k^*}^*)$ for each of the configuration. The buildings which are covered are shown in green, whereas those not covered are in red.

We observe that Config. 1 is only able to cover buildings in close proximity to the antenna, as it is an undirected radiator with no beamforming capability. The radiated power therefore spreads isotropically, resulting in rapid spatial decay and limited reach across the campus.

Configs. 2 and 3 exhibit significantly improved performance due to their increased directivity. By forming a focused main lobe, these configurations concentrate energy into a narrower angular region, allowing coverage of a larger portion of the campus. Config. 3 slightly outperforms Config. 2 because the applied steering introduces a controlled phase shift that reorients the main beam towards the south-east region, enabling coverage of one additional building.

Config. 4, despite having a highly directional planar aperture, performs poorly in this scenario. Without phase steering, its main beam remains fixed at broadside, which does not coincide with the spatial distribution of building clusters. As a result, much of the radiated energy is directed away from the regions of interest, leading to inefficient coverage despite its high inherent directivity.

Finally, Config. 5 achieves the best overall performance. With optimally chosen phase shifts, the main lobe is steered directly towards the densest cluster of buildings in the western region of the campus. In addition, the formation of a pronounced sidelobe in this direction further enhances coverage of peripheral structures within the cluster. This demonstrates that, in realistic propagation environments, controlled sidelobe placement can contribute positively to coverage efficiency when aligned with spatial demand.

Overall, these results highlight that high directivity alone is insufficient for optimal performance; instead, the combination of beam steering and spatial alignment with user distribution is essential for maximising the combined objective function.

4.5 Discussion

The results presented in Sections 4.1–4.4 are in good quantitative agreement with the theoretical predictions derived in Section 2, validating both the physical model and the numerical implementation.

The most fundamental assumption underlying all five configurations is the *separability approximation*: mutual coupling between antenna elements is neglected, so the total field is treated as a coherent superposition of independent radiators. In practice, neighbouring elements perturb each other's current distribution, modifying both the element pattern $g(\vartheta)$ and the array factor. This effect becomes significant for $d < \lambda/2$ and partially accounts for the broadened main lobe observed at $d = \lambda/4$ in Figure 2, beyond the aperture.

A second limitation is the assumption of *free-space propagation*: the link budget $S_i \propto G^{(k)}/r_i^2$ used in the campus optimisation ignores reflections, diffraction, and shadowing by buildings. In a real campus environment, multipath propagation would modify the received power at each building, potentially covering some buildings that appear uncovered in our model and vice versa. A more realistic model would incorporate a path-loss exponent $n > 2$ and building attenuation factors.

The *sinusoidal current distribution* assumed for the half-wave dipole is a further approximation, valid for thin wires but increasingly inaccurate for thick elements or elements in close proximity to neighbours. This introduces a small systematic error in the element factor $g(\vartheta)$, which propagates into the directivity computation, consistent with the discrepancy observed between the simulated and theoretical planar array directivity.

Finally, the *gradient ascent optimisation* of the campus coverage objective is not guaranteed to find the global maximum of $J_k(\boldsymbol{\delta})$, as the sigmoid smoothing introduces a non-convex landscape. The homotopy continuation mitigates this by progressively sharpening the objective, but sensitivity to the initialisation $\boldsymbol{\delta}^{(0)} = \mathbf{0}$ cannot be entirely ruled out. A more robust approach would use multiple random initialisations or a global optimisation method such as simulated annealing.

Despite these limitations, the simulation correctly reproduces all qualitative features predicted by theory: beamwidth narrowing with N , the universal -13.3 dB sidelobe level, grating lobe formation at $d = \lambda$, beam squint scaling as $1/\sin \vartheta_0$, and the directivity advantage of planar over linear arrays.

5 Conclusions

This project set out to determine how the geometry and phase control of half-wave dipole arrays shape the radiated far field, and to apply the resulting design rules to wireless coverage of the École Polytechnique campus. Five configurations were studied, a single dipole, broadside and steered linear arrays, and broadside and steered planar arrays, using the pattern multiplication rule $\tilde{U} = g(\vartheta) \cdot |\text{AF}|^2$ to separate dipole physics from array geometry, and a smoothed coverage–efficiency objective J_k optimised by gradient ascent with homotopy continuation on the sigmoid sharpness α .

The principal finding is that the textbook scaling laws of array antenna theory hold quantitatively in our implementation, and that they translate directly into engineering design rules. Directivity grows linearly with the number of elements ($D^{(2)} \approx N$ and $D^{(4)} \approx \pi N_x N_z / 2$), agreeing with theory to within 0.2 dB for linear arrays and 0.8 dB for the planar array. The half-power beamwidth of the broadside ULA narrows as $0.886 \lambda / (Nd)$, the first sidelobe level locks at the universal -13.3 dB value of uniform amplitude weighting, and grating lobes appear precisely at $d = \lambda$ in the predicted directions, confirming $d = \lambda/2$ as the canonical grating-lobe-free choice. Steering away from broadside costs directivity as $\sin \vartheta_0$, with usable performance maintained inside a $\pm 60^\circ$ envelope before scan loss becomes prohibitive. Applied to the campus, the steered planar array (Config. 5) wins decisively, covering 44 of 60 buildings with $J_5 = 0.8607$ against 10/60 for the single dipole and 9/60 for the unsteered planar array — the latter result, perhaps counter-intuitive at first sight, illustrating that high directivity is worthless if it is pointed in the wrong direction.

The broader message is that beamforming is fundamentally a problem of *matching* the radiation pattern to the spatial distribution of receivers, not merely a problem of generating the narrowest possible beam. The unsteered planar array, despite having the highest raw directivity of the five configurations, performs worse than a single dipole on the campus problem because its beam is fixed at broadside and the buildings are not. The steered linear array (Config. 3) is only marginally better than its broadside counterpart (41/60 vs. 40/60) because a single steering parameter cannot simultaneously align the beam with multiple building clusters. Only when the planar array is allowed to steer in two dimensions does the combination of high directivity and angular flexibility produce a substantial gain over simpler configurations. Equally instructive is the observation, visible in Figure 9, that one of the optimised sidelobes of Config. 5 lands on a peripheral cluster of buildings: under a coverage-driven objective, the optimiser learns to use sidelobes constructively rather than treat them purely as waste.

The results are subject to several limitations that are worth stating clearly. The separability approximation neglects mutual coupling between elements, which becomes significant at sub-half-wavelength spacing and partly accounts for the broadened main lobe seen at $d = \lambda/4$ beyond what aperture reduction alone predicts; a full-wave electromagnetic solver would be needed to capture this quantitatively. The free-space link budget $S_i \propto G^{(k)} / r_i^2$ ignores reflection, diffraction, and shadowing, all of which matter on a real campus with concrete buildings, and a realistic urban model with path-loss exponent $n > 2$ and explicit attenuation per building would likely shift the optimal phase configuration. The sinusoidal current distribution on the dipole is itself an idealisation valid for thin wires only, and we suspect this contributes to the residual 0.8 dB gap between simulated and theoretical planar directivity. Finally, gradient ascent on a sigmoid-smoothed non-convex objective is not guaranteed to find the global maximum, even with homotopy continuation; the solution we report is the best local optimum reached from $\delta^{(0)} = \mathbf{0}$, and a different initialisation could in principle do better.

For students working on a similar project, two practical recommendations emerge from our experience. First, derive the array factor in closed form before writing any simulation code: the geometric series $\sum_n e^{in\psi}$ collapses to $\sin(N\psi/2) / \sin(\psi/2)$ in two lines, and having the analytical form available makes every numerical result interpretable rather than mysterious. Second, validate the simulation on cases with known answers, the single dipole's 2.15 dBi, the universal -13.3 dB sidelobe level, the $\pi N_x N_z / 2$ planar directivity, before trusting it on the full campus problem; we caught more than one indexing error this way.

Contributions

Villads Eskjaer and Clarisse Aubert organized the project. Before writing the report, all group members collectively edited a rough draft on a Latex Document before it was formally written in this final report. Villads Eskjaer and Camille Travers created the two algorithms (general analysis algorithm and applied to campus) and implemented it on computer. Panagiotis Lioultas worked on the theoretical background of this project while Clarisse Aubert finalized editorial writing. Camille Travers and Villads Eskjaer carried out the simulations. All authors analysed the data and discussed the results. Clarisse Aubert and Panagiotis Lioultas wrote the introduction and conclusion with input from all members after collectively rereading and going through the report.

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